

to $\frac{\text{LMI}}{\text{MPI}} = 0.8$ (not shown), the distribution can no longer be considered uniform after this upper value. Therefore, we choose to integrate the bimodal aspect of the distribution in our model by fitting the overall $\frac{\text{LMI}}{\text{MPI}}$ distribution to two mixed log-normal distributions. The first lognormal probability density,

$$f_{\log}\left(\frac{x - l_1}{s_1}; \mu_1; \sigma_1\right)/s_1$$

weights $w_1 = 0.55$ with parameters given by $\sigma_1 = 0.18$, $s_1 = \exp \mu_1 = 0.65$, $l_1 = -0.014$, while the second lognormal probability density

$$f_{\log}\left(\frac{x - l_2}{s_2}; \mu_2; \sigma_2\right)/s_2$$

weights $w_2 = 0.45$ with parameters given by $\sigma_2 = 0.15$, $s = \exp \mu_2 = 0.96$, $l_2 = -0.02$.

(Again, not shown) It is also observed that the distribution adopts the same visual characteristics in the East Pacific, South Pacific, West Pacific and South Indian basins, namely one peak between 0.6 and 0.7 and another between 0.85 and 1, with a non-negligible tail, supporting a form of universality of the distribution. Table 1 supports this statement by showing the mean and the weight w of the two mixed distributions in the four basins.

Basin	East Pacific	South Pacific	West Pacific	South Indian
f_1 Parameters	$\exp \mu_1 = 0.70$ $w_1 = 0.57$	$\exp \mu_1 = 0.63$ $w_1 = 0.64$	$\exp \mu_1 = 0.68$ $w_1 = 0.60$	$\exp \mu_1 = 0.61$ $w_1 = 0.48$
f_2 Parameters	$\exp \mu_2 = 1.01$ $w_2 = 0.43$	$\exp \mu_2 = 0.97$ $w_1 = 0.36$	$\exp \mu_2 = 0.89$ $w_2 = 0.40$	$\exp \mu_2 = 0.87$ $w_2 = 0.52$

Table 1: Mean and weight of the two mixed log-normal distributions f_1 and f_2 in four relevant basins. Associated to 392 storm cases in the EP, 164 in the SP, 593 in the WP and 334 in the SI basins.

However, this universality is visually broken in the North Atlantic and North Indian oceans, where the ratio looks more randomly distributed. The random aspect of this shape could be attributed to the lack of MPI data for those two basins, making their distributions possibly inaccurate. Indeed in the North Atlantic, the distribution was associated with 146 storm cases, whereas 55 points were available in the North Indian.

All these findings add a nuance to what is usually taken for granted, namely that a storm has nearly the same probability of reaching any fraction of its MPI [Emanuel, 2000] and show the relevance of conducting a deeper analysis of the LMI.

We take the analysis of the bimodal shape a step further by exploring the dependency between the $\frac{\text{LMI}}{\text{MPI}}$ ratio and the distance separating the storm's LMI location and the nearest coast. Indeed, as the storm approaches land, the environmental conditions are less favourable to the storm's intensification process, which can significantly impact the value of its lifetime intensity in this region. Figures 4a 4b and 4c explore this aspect in more detail. By showing the distribution shape for three different regions from the coast,

one can observe that the closer the storm is to land, the lower chance it has to reach a large fraction of its MPI and hence intensify significantly.

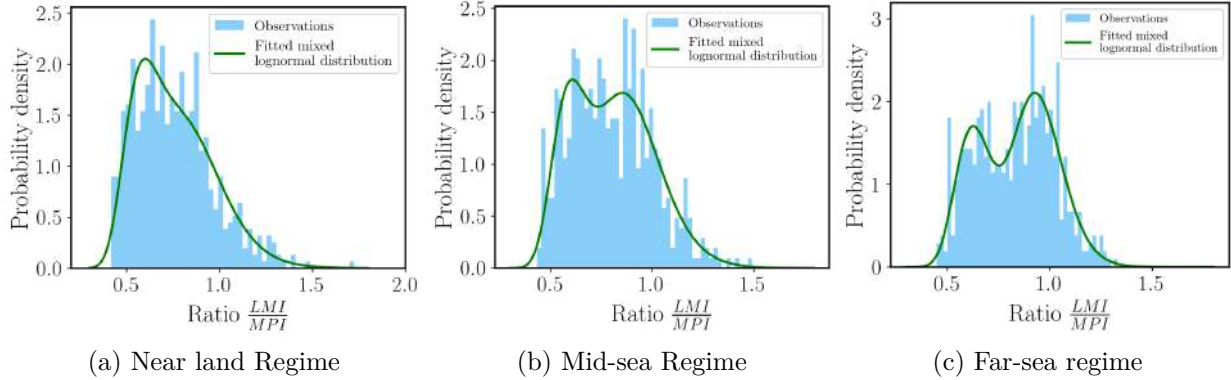


Figure 4: Probability distribution of the $\frac{LMI}{MPI}$ distribution for three different regimes. The entire set is separated into three different regions corresponding to different ranges of distances between the LMI and the nearest coastal point. In (a) associated with 597 storm cases for which the distance is below 300 km, in (b) associated with 491 storm cases for which the distance is between 300 and 800 km and (c) associated with 596 storm cases for which the distance is above 800 km. The fitted distributions are also displayed for each regime. Statistical tests do not reject the hypothesis that the observed distributions follow a mixed lognormal distribution weighted differently depending on the three different cases ($p_0 > 0.05$).

Looking at these Figures, it seems that the storm's distance from land controls the bimodal distribution shape and the weight of each log-normal distribution. In light of the mentioned physical phenomena responsible for the bimodal shape, the vertical shear of the horizontal wind seems to be particularly significant near the coast. In contrast, its impact decreases as the storm evolves further and further over the ocean. Similarly, the sea's response to the storm follows the opposite effect; the importance of this phenomenon seems all the more important as the storm evolves far from the coast. Those findings are not surprising. Indeed as the storm approaches land a portion of itself evolves over land, the environmental conditions become less and less favourable to high wind speed values (more drag, less moist surface fluxes) making the cyclone even more sensitive to the vertical wind shear [Li-na and Yuan, 2016]. On the contrary, it is far from the coast over the ocean that the conditions for their intensification are the most favourable. Thus, the storms are more susceptible to having reached high lifetime maximum intensities and are more affected now by the second phenomenon, namely the ocean's response to the hurricane.

In light of these considerations, I tried to fit three mixed log-normal distributions corresponding to the three mentioned range of distances.

The first mixed log-normal probability density accounting for the ratio's near-land behaviour weights 0.82 with parameters given by $\sigma_1 = 0.24$, $s_1 = \exp \mu_1 = 0.66$, $l_1 = 0.02$, while the second log-normal probability density weights 0.18 with parameters given by $\sigma_2 = 0.18$, $s = \exp \mu_2 = 0.99$, $l_2 = 0$.

Secondly, the first mixed log-normal probability density accounting for the ratio's behaviour between 300 and 800 km weights 0.53 $\sigma_1 = 0.16$ with parameters given by $s_1 = \exp \mu_1 = 0.57$ and $l_1 = 0.03$, while the second log-normal probability density weights 0.47 with parameters given by $\sigma_2 = 0.17$, $s = \exp \mu_2 = 0.91$, $l_2 = -0.01$.

Thirdly, the first mixed log-normal probability density accounting for the ratio's far from the coast weights 0.45 $\sigma_1 = 0.17$, with parameters $s = \exp \mu_1 = 0.64$ and $l_1 = 0.01$, while the second log-normal probability density weights 0.55 with parameters given by $\sigma_2 = 0.12$, $s = \exp \mu_2 = 0.93$, $l_2 = 0.03$.

Therefore, we fail here to highlight two mixed log-normal perfectly weighted as a function of the distance between the storm's LMI location and the coast. Indeed, even though the first mixed log-normal distribution's weight decreases continuously from the first case to the third case, the peaks of the two distributions are different and so are their widths (from their standard deviation). Although a perfect relation could exist and be brought to light, it could also be that the value of the peaks depend on distance too, or that fitting a universal distribution in such a way is impossible and a deeper analysis per basin and season may be required. However, the dependence has still been clearly highlighted and the impact of the storm's LMI distance from the coast on its estimation is clear. These findings are particularly significant in the context of the development of our model. In particular, distinguishing the distribution of the ratio between the three regimes enables a better estimation of the LMI near the coast, where the storms are the most likely to make landfall and cause disasters. We will hence use, as a first approximation, three distributions in a discrete way in the model. The first distribution is used to simulate the cases where the effects of the coast on the TCs are strong, the second when the storm is moderately affected by those effects, and the third one when the storm is in the middle of the ocean, without any interactions with land.

Finally, even though we will consider those distributions universal as a first approach to the problem, it is too early to say if they indeed are, namely if the ratio displays the same behaviour in any basin and any month. The ratio seems universal in four different basins as discussed above (and still it is only a visual confirmation, the weights of the two mixed distributions vary from different basins and so do their peaks, which considerably affects their characteristics and the proportion of the ratio) but how the seasonal variations affect this shape is not highlighted and hence not clear. It is also not yet clear if the bimodal shape is universally ruled by the distance, or if one can expect significant differences when studying this dependence locally. Indeed, a recent study [Ge et al., 2018] highlights the monthly variations of the intensification process. It is revealed that during the late fall season, the TCs' thermodynamic efficiencies are greater and the subsurface ocean conditions favour a weaker cooling effect (potentially controlling the value of the second peak of the mixed log-normal distribution given our assumptions).

We will discuss the sensitivity of the ratio $\frac{\text{LMI}}{\text{MPI}}$ distribution in a later section, and will more specifically try to estimate the impact of all the assumptions made in this section on the model's results.

4.1.3 Wind speed profile at landfall

To motivate the choice to generate, from its distribution, a variable (R_{34} or RMW) over the other in Equation (6), we need to display the correlation between α and the two variables and select the one with the lowest correlation. Figure 5a and 5b illustrate the dependency between α and the two variable R_{34} and RMW.

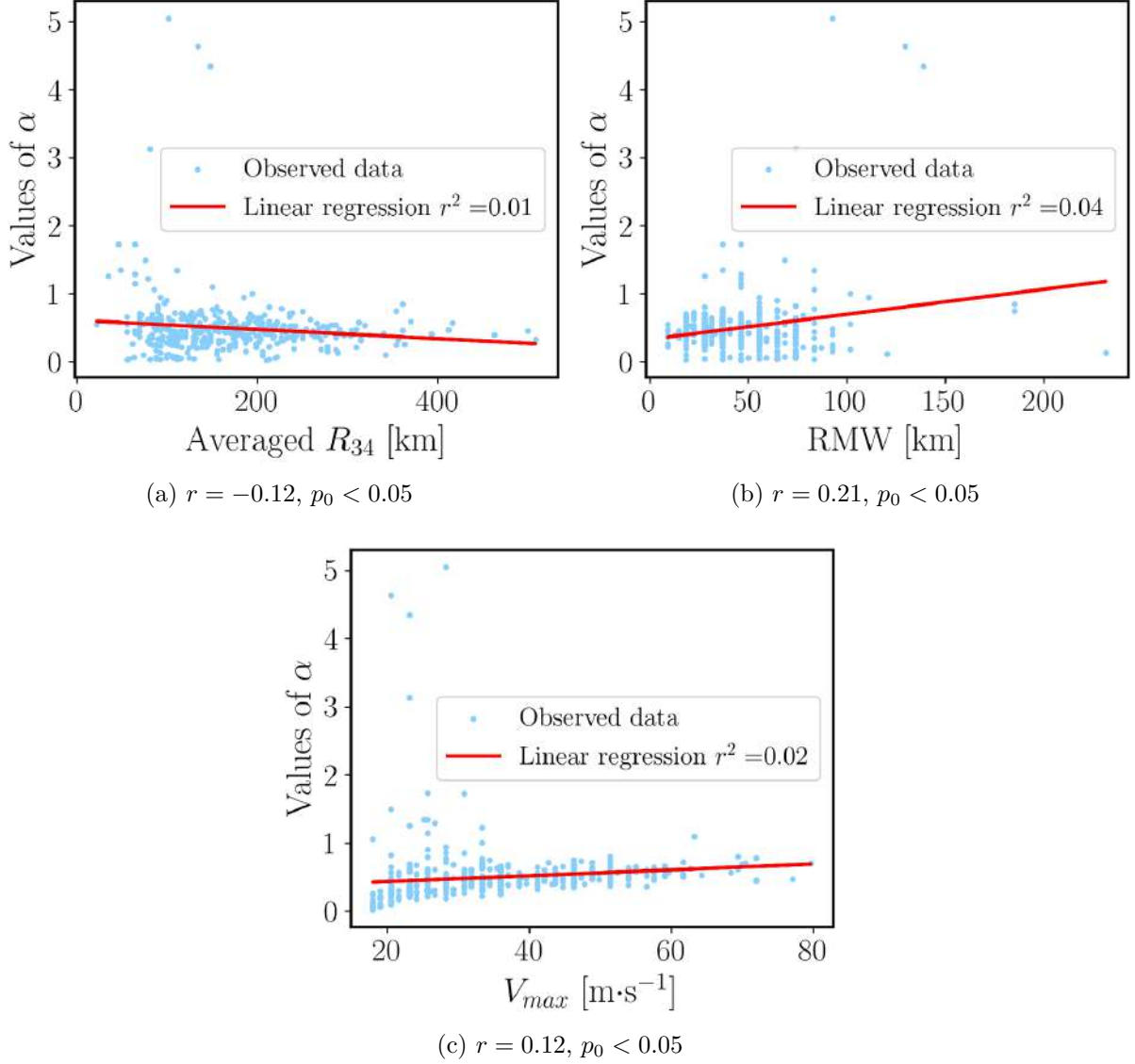


Figure 5: A scatterplot depicting the relationship between the decay exponent α and the averaged radius of 34 knots wind speed R_{34} in (a), the radius of maximum wind RMW in (b) and the maximum wind speed V_{max} in (c), associated with 404 observed storm cases. The Pearson correlation coefficients displayed are $r = 0.12$ in (a), $r = 0.21$ in (b) and $r = -0.12$ in (c) and the p -values are both under the significance level.

The computed Pearson correlation coefficients and the p_0 values suggest that the dependence of α on R_{34} and RMW are weak, but nonetheless significant. However, as RMW

shows a slightly better correlation and could not be fitted to any commonly known distribution, taking R_{34} as the second estimated parameter is a good choice. Moreover the absence of statistical relationship between the maximum wind speed at landfall V_{max} and the decay exponent α ($r^2 = 0.02$, $p_0 < 0.05$) is demonstrated by the plot in Figure 5c. This result does not go against what has been highlighted in the literature [Mallen et al., 2005], suggesting a tendency of increasing α with increasing V_{max} . Indeed, this study focused on the observed wind profile globally without distinguishing the behaviour at landfall and above the ocean. Therefore, considering that TCs' physics over land may considerably change from above the ocean, it is impossible to conclude that this empirical dependency should be seen within our framework. However, a moderate statistical relation is observed between V_{max} and R_{34} , as shown in Figure 6.

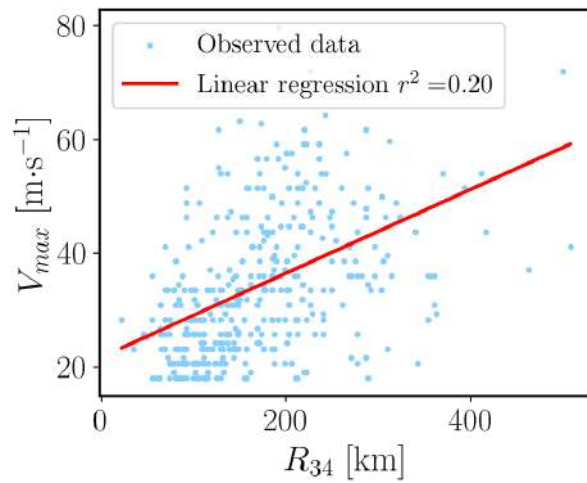


Figure 6: A scatterplot depicting the relationship between the averaged maximum surface wind speed V_{max} and the averaged radius of 34 knots winds R_{34} , associated with 404 observed storm cases. The Pearson correlation coefficient displayed is $r = 0.44$ and the p -value is under the significance level ($p_0 < 0.05$).

In the end, these results indicate that, firstly, α may be chosen entirely independently from all the variables, confirming the relevance of generating new wind speed profiles in such a manner. Secondly, it is worth noting that V_{max} and RMW moderately correlate when choosing the value of RMW from its distribution. Depending on the precision we want to include in the model, we might want to consider this dependence when implementing the wind speed profile.

Having this analysis in mind, Figure 7 displays the α and Averaged R_{34} distributions with different commonly used fitting distributions. K-S test does not reject the hypothesis that α follows a Burr distribution (distribution part of the log-logistic distributions) with shape parameters $c = 4.83$ and $d = 0.49$, and scale $s = 0.60$. However, all the other usual distributions are rejected by the test. Especially, the exponent α cannot be described using a Log-normal distribution.

Furthermore, K-S test does not reject the hypothesis that the averaged R_{34} can be described either by a lognormal distribution with parameter sigma $\sigma = 0.49$ and scale $s = \exp \mu = 80.94$ or a Beta distribution with parameter $a = 4.56$, $b = 5240.63$ and scale

$s = 104237.26$.

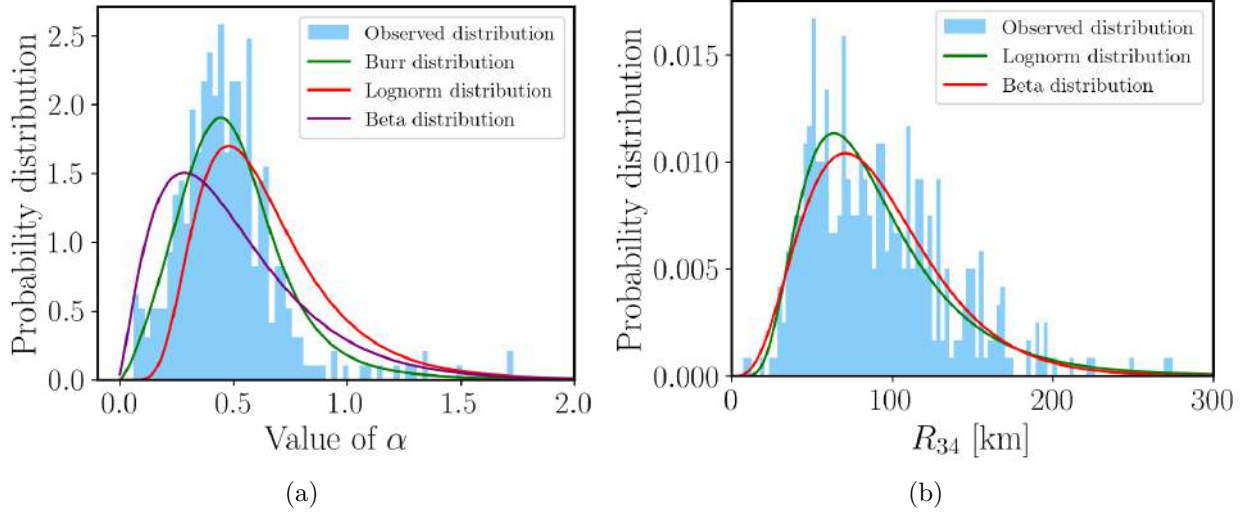


Figure 7: Probability distributions of the decay exponent α in (a) and the averaged Radius of 34 knots winds R_{34} in (b), associated with 404 storm cases. For visualisation purposes, conventional fitting distributions are displayed. Statistical test does not reject the Burr distribution in (a) and both Lognormal and Beta distributions in (b) ($p_0 > 0.05$).

4.1.4 Summary of the extracted distributions

Given the difficulty of making precise meteorological measurements and the computational errors (especially in computing the power-law exponent of the wind profile α and the decay parameter κ), it is too early to tell if the displayed distributions are, in reality, universal and following the fitted distributions. In the context of our model's implementation, we only use them as a helpful tool to generate new realistic values and incorporate some variations in their occurrences. Some care should be then taken when interpreting their behaviour in terms of the underlying physics. I attempt to explore the underlying physical phenomenon in the case of the $\frac{LMI}{MPI}$ ratio but the fact that a non-usual Burr distribution approximates well the exponent α does not necessarily entail that the parameter follows the latter and hence describes a non-usual physical process. The distributions are so far only applicable within the framework of our model's implementation. All the according fitted distributions's characteristics are summarised in Tables 2, 3 and 4 in the appendix section A.

4.2 Averaged-rain rate dependencies

This section aims to identify the maximum averaged rain rate dependencies. Figure 8 depicts the empirical relationships between the logarithm of the maximum averaged rain rate and the logarithm of relevant quantities: maximum surface winds V_{max} , the water vapour pressure e_s and the averaged terrain height $\langle H \rangle$ within a 200-km radius around the landfall's eye location. All the computed Pearson correlation coefficients indicate no clear relationship between the averaged rain rate and the mentioned quantities.

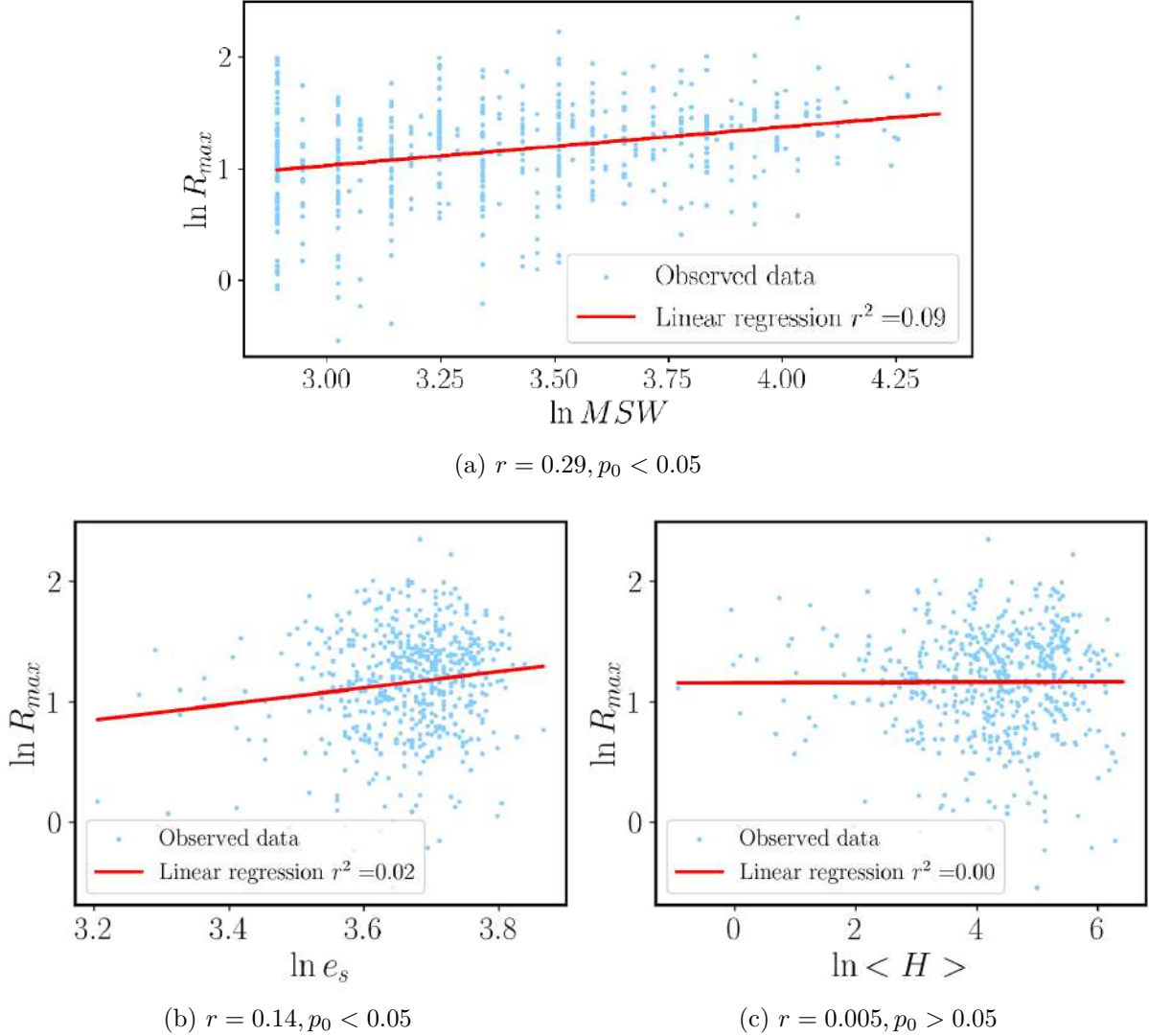


Figure 8: Scatterplots depicting the relationships between the natural logarithm of maximum averaged azimuthal rain rate R_{max} and the natural logarithms of the maximum sustained winds MSW at radius of maximum wind speeds RMW in (a), the saturated water vapour pressure e_s in (b) and the averaged height $<H>$ in (c). The Pearson correlation coefficient displayed are $r = 0.29$ in (a), $r = 0.14$ in (b) and $r = 0.005$. The p -value is under the significance level ($p_0 < 0.05$) in (a) and (b) and above ($p_0 > 0.05$) in (c).

Those findings reveal that the relation between a TC's intensity as well as the saturated water vapour pressure and its precipitations at landfall are weak when studying the dependence worldwide. This suggests that weak hurricanes in the West Pacific could show more precipitations than intense TCs in the East Pacific for example and that saturated water vapour pressure.

Moreover, even though the p -value computed from the statistical relationship between the rain rate and the averaged height is below the significance level, the most probable scenario is that the two variables at most weakly correlate. Therefore, at landfall, the

average height of the coast where the TC makes landfall does not seem to affect the TCs' precipitations in any way. This link might depend on specific conditions, and studying it worldwide is not an ideal way to highlight their relationship. Indeed, it is also essential to keep in mind that the associated increased roughness and friction of the mountainous surface can also lead the storm to a quicker dissipation [Chen et al., 2010] and hence have the opposite effect. It might also be that precipitations in the coastal area are not as affected by topography as when the storm evolves further over land. Indeed, many complex mechanisms are in play on the coast as TCs change their thermodynamics regime when shifting from ocean to earth. Moreover (not shown), no better correlations were found between the rain rate and the maximum height ($r = 0.05$), the standard deviation of the averaged height ($r = 0.02$) and the radius of maximum winds RMW ($r = 0.1$).

These findings go against what we had expected and illustrate the complexity of the rain mechanism at landfall and the difficulty of highlighting worldwide, universal relationships. Regarding the dependence between the rainfall and the maximum wind speed at landfall, it might be relevant, in further studies, to restrict the analysis to the different basins and reduce, in such a way, the environmental variability. Indeed specific rain features are usually studied by regions, as they are highly dependant on local weather patterns and remain still difficult to extract [Breña-Naranjo et al., 2015, Ye and Fang, 2018, Gori et al., 2022].

However, if one has access to such data, a study of the dependence between the TCs' axisymmetric rainfall rates, or their vertical wind shears, and the rainfall intensities, could show a clearer linear regression and be eventually implemented in the context of our model [Cheung et al., 2018]. Instead of focusing on the averaged maximum azimuthal rain rate, one could also study this dependence distinguishing between the four TC quadrants. In particular the right front quadrant as they are the most affected by strong mesoscales convective systems shown to produce heavy rainfall. This analysis could show better results as most research reviews emphasises on the asymmetric distribution of TCs' rainfall [Chen et al., 2010, Cheung et al., 2018].

Therefore, given the lack of direct relationships between the averaged rain rate and the studied quantities, it is unfounded to fit any coefficients to the multiple linear regressions and extract any universal power-law decays. The model will hence not integrate any rainfall predictions so far. If we want to further integrate them, the study will have to be done per basins, or at least separate between the wet basins (WP, NA) and dry basins (EP, SI). This could show promising results as the differences in TC precipitations between basins are in the same order of magnitude to the difference between intensity changes [Lonfat et al., 2004]. More variables may also need to be considered such as the TC size, as it has also recently showed to play a major role [Lavender and McBride, 2021], even if this study confirms the difficulty of isolating all the inter-related factors at play in TCs' rainfall mechanism.

4.3 Testing the model

Now that we have extracted the κ and the $\frac{LMI}{MPI}$ ratio distributions, we can generate new storms with intensities decaying differently from the parent cyclones in order to account for environmental changes. This section first visually presents a typical decay implemen-

tation for a specific storm to highlight all the potential decay scenarios. Then, the model simulates a 100 times 40-year block of new synthetic cyclones. The return periods in major cities are presented to estimate the model's goodness in predicting realistic TCs' intensity values at landfall.

4.3.1 Decay implementation

For visualisation purposes, Figure 9 depicts the spread of the intensity decays of a hundred synthetic storms, generated from an example parent cyclone in the dataset. It also displays the fit to Equation (4) of the parent storm's decay. This figure illustrates how even though the intensification process is not allowed by the model, it can still brilliantly approximate the wind speed value at landfall. This is the ultimate aim of the model since it needs to evaluate the risk in vulnerable areas. This figure also illustrates the decay implementation in the two different environment, over land and the ocean (separated by the horizontal line), and shows the difference in the sharpness of their decay.

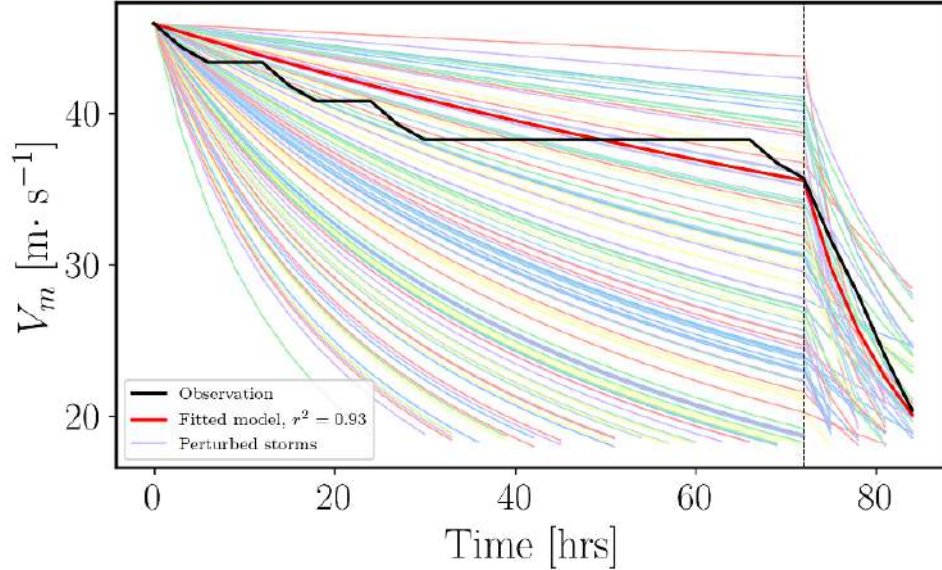


Figure 9: Maximum wind speed of the storms as a function of time. The left zone corresponds to the regime where the storms are over sea, while the right zone corresponds to the regime where the storms are over land. The observed storm is JUDY which happened in the year 1982 in the West Pacific. The built model shows how Equation (4) can indeed approximate the trend of the observation given a fitted $\kappa_{\text{land}} = 2.45 \cdot 10^{-8} \text{m}^{-1}$ over the ocean and $\kappa_{\text{land}} = 5 \cdot 10^{-7} \text{m}^{-1}$ over land. Given the extracted κ -distributions distinguishing between the ocean and the land behaviour, the model simulated 100 more storms from the observed LMI, showing the range of plausible intensity decays.

4.3.2 Return periods

This part compares the TCs' characteristics at landfall obtained from the model to the observed ones. We calculate different return periods curves as a function of landfall intensity in several cities exposed to cyclones. By applying this model, which does not

consider TCs' regional and seasonal variations, to specific areas, we can estimate how reasonable the assumptions for developing a worldwide model are. We also present the extended 80 years-return period, useful to assess the TCs risk in this century if the climate conditions remain the same. The model is accepted as a good predictor of the TCs' intensity in the area if the observed return period falls within 95% of the simulated return period's spread. Indeed, if each recorded observed point lie within the range, then the hypothesis that the model approximates well the situation in the area cannot be rejected at the 5% significance level. These lower 5% and upper 95% confidence bounds will be displayed for each plotted return period. Its estimation is accurate for the first 40 years and useful to compare the observations to the simulations, but its prediction, when extending further, may not be as accurate. In particular, this is because we resample the same ensemble to extend the period, which is not ideal. More rigorous ways of doing it are usually implemented, in particular by fitting extreme values distributions [Kotz and Nadarajah, 2000], but could not be conducted here due to the lack of time at the end of the project.

West Pacific Basin

This paragraph explores the return periods displayed by the simulated storms within a 200km radius around four major cities in the West Pacific basin: Shanghai and Hong-Kong in Figure 10 and Tokyo and Manila in Figure 11. Those figures also depicts the observed return periods in the according areas.

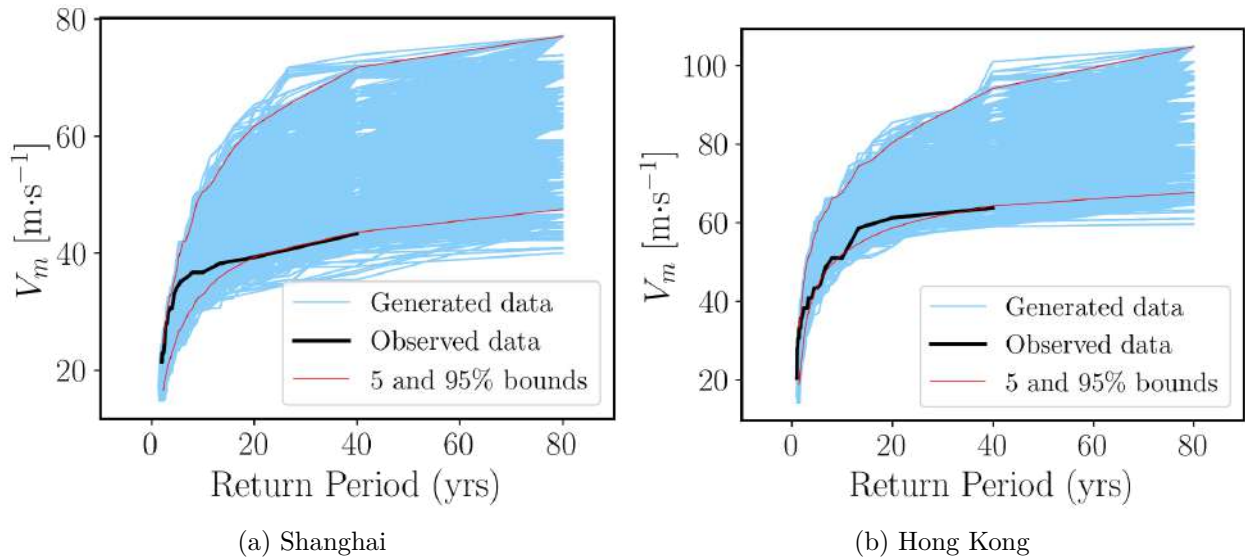


Figure 10: Return period curves of landfall intensity V_m within a 200km radius around (a) Shanghai and (b) Hong Kong. The simulated return periods are associated with hundred 40-year blocks, while the observed return period is associated to the past 40 years records in the two regions. The 40-years simulated blocks are re sampled and concatenated to extract the 80 years return periods. The observed return periods both lie within the 95% range of simulations.

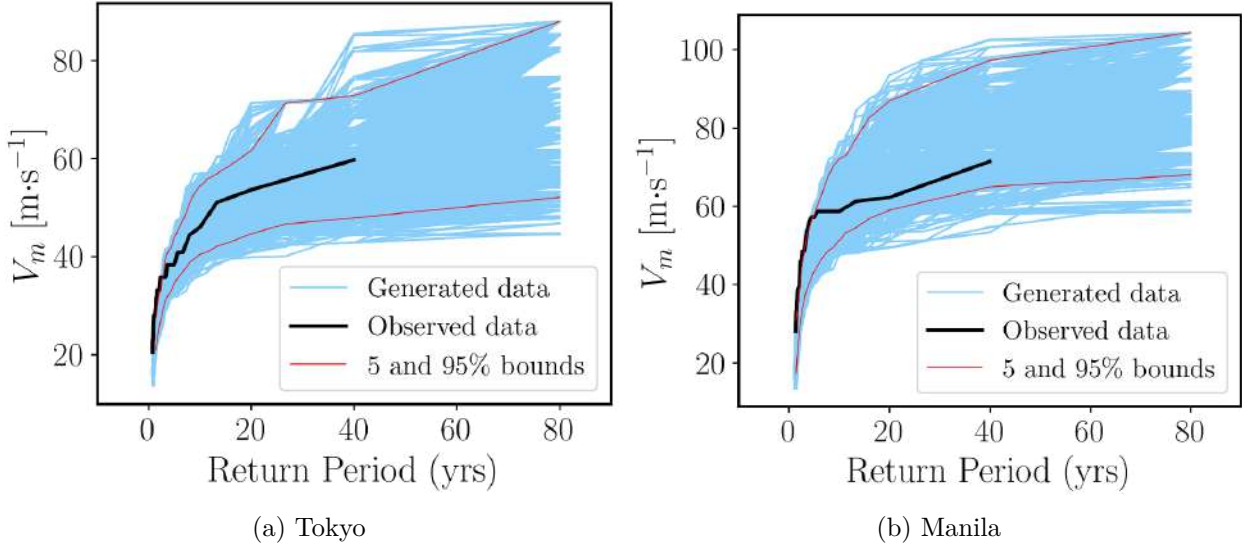


Figure 11: Return period curves of landfall intensity V_m within a 200 km radius around (a) Tokyo and (b) Manila. The simulated return periods are associated with hundred 40-year blocks, while the observed return period is associated to the past 40 years records in the two regions. The 40-years simulated blocks are re sampled and concatenated to extract the 80 years return periods. The observed return periods both lie within the 95% range of simulations.

These figures show how the model brilliantly predicts the TCs' activity around the main exposed cities of the West Pacific ocean. Indeed, all the observed return period curves lie within the confidence ensemble spread provided by the model. Both the shape of the curves (related to a good estimation of the intensity) and the location of the curves (related to a good estimation of the frequency of events) are predicted by the model.

Moreover, the return period curves show that the probabilities of a particular typhoon event vary from city to city, with Manila and Hong Kong being the most exposed to TC hazards. In particular, it is interesting to note that the scenario where Hong Kong and Manila host devastating events in the century with wind speeds above $350 \text{ km} \cdot \text{hr}^{-1}$ is a prediction of the model.

East Pacific Basin

This paragraph explores the return period displayed by the simulated storms within a 400 km radius around on major city of the East Pacific basin: Hawaii in Figure 13. A higher radius is chosen here as less data was available around the studied location. This figure also depicts the observed return period curve in the area. It shows that the observed return period is mostly located below the upper boundary of the significance level except for the $50 \text{ m} \cdot \text{s}^{-1}$ intensity event which is under estimated by the model.

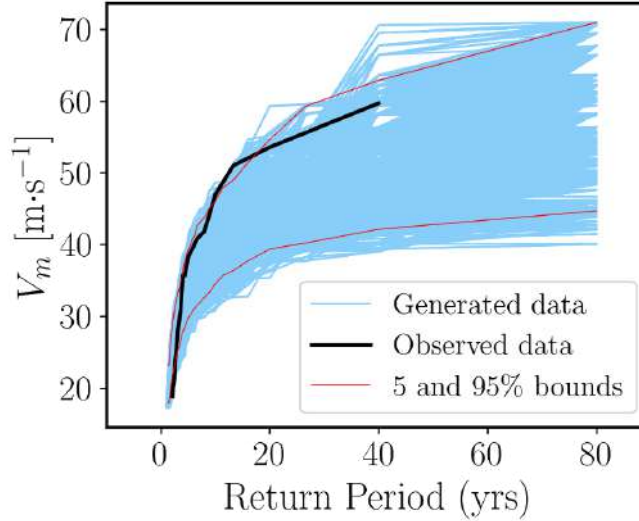


Figure 12: Return period curves of landfall intensity V_m within a 400 km radius around Hawaii in the East Pacific basin. The simulated return periods are associated with hundred 40-year blocks, while the observed return period is associated to the past 40 years records. The 40-years simulated blocks are re sampled and concatenated to extract the 80 years return periods. The observed return period mostly lies within the 95% range of simulations.

North Atlantic and North Indian basins

We will now explore the return periods displayed by the simulated storms within a 400 km radius around two major cities of the North Atlantic basin: Miami and New Orleans in Figure 13.

These figures also depict the observed return periods in the according areas. Firstly, the observed return period in Miami is clearly under estimated by the model. Indeed, except for the intensity events' around $80 \text{ m}\cdot\text{s}^{-1}$, each observed event is above the upper confidence bound. Secondly, the frequency of mid wind speed events (below $60 \text{ m}\cdot\text{s}^{-1}$) in New Orleans is also slightly under estimated by the model, while stronger wind speed events are well estimated.

In light of these considerations, in its current implementation, the model overall fails to predict the TC activity in those two exposed cities of the North Atlantic.

This paragraph also explores the return period displayed by the simulated storms within a 400 km radius around Dhaka in in Figure 14.

This city is one of the most exposed cities to cyclones activity in the North Indian basin. This figure also depicts the observed return period curve in the area. It shows that the observed return period is located between the range of confidence of the simulations, giving confidence in the model's predictions of TC activity around Dhaka.

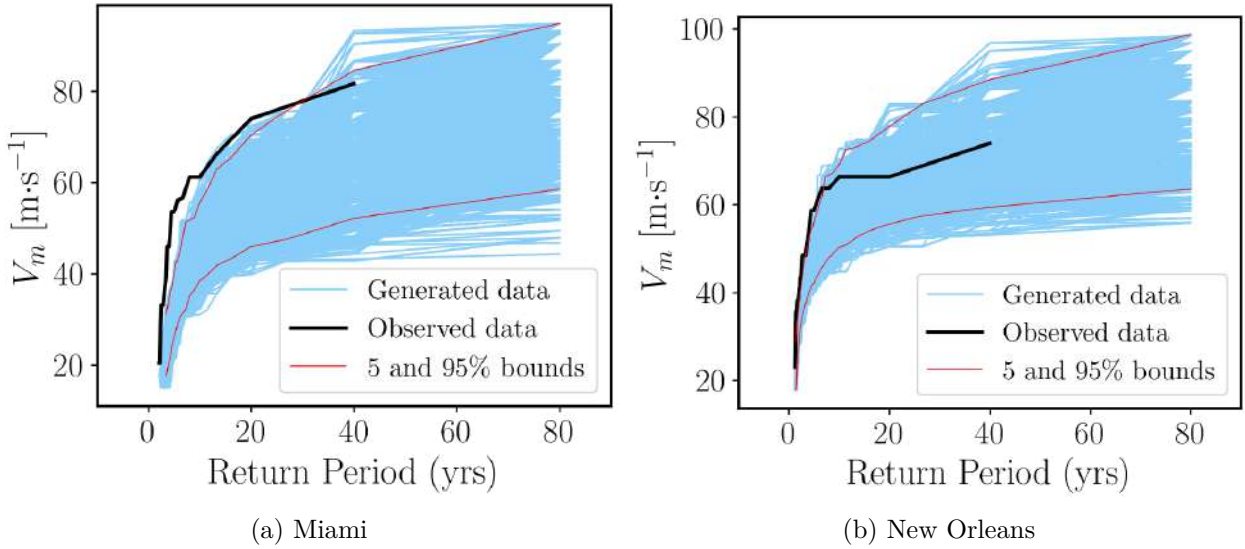


Figure 13: Return period curves of landfall intensity V_m within a 400 km radius around (a) Miami and (b) New Orleans in the North Atlantic basins. The simulated return periods are associated with hundred 40-year blocks, while the observed return period is associated to the past 40 years records in the two regions. The 40-years simulated blocks are re sampled and concatenated to extract the 80 years return periods. The observed curves fail to always fall within the spread of confidence.

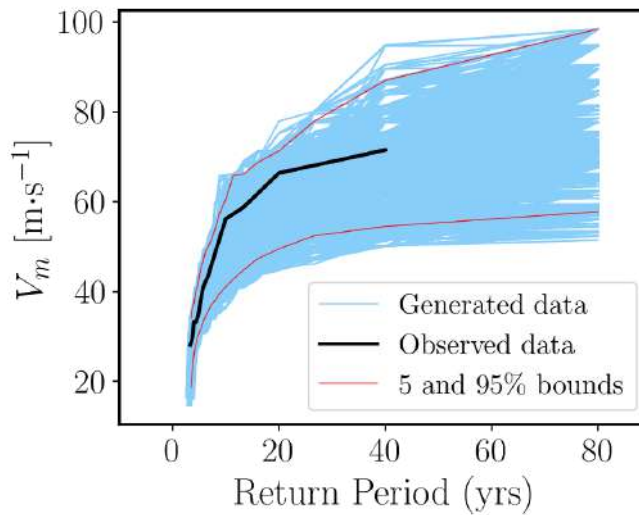


Figure 14: Return period curves of landfall intensity V_m within a 400 km radius around Dhaka in the North Indian basin. The simulated return periods are associated with hundred 40-year blocks, while the observed return period is associated to the past 40 years records. The 40-years simulated blocks are re sampled and concatenated to extract the 80 years return periods. The observed return period lie within the 95% range of simulations.

Overall, even though the TC activity seems well approximated by the model in a few major exposed cities of the West Pacific and North Indian, it fails to predict their behaviour in the North Atlantic and the predicted behaviour in Hawaii is questionable. However, we are not short of reasons to explain these deviations, and many easily implementable improvements are possible.

Firstly, the model assumes that cyclones start to feel the effect of land only when their eye crosses the ground, so when half of the storm is already over land. As discussed when investigating the decay parameter κ distribution, the effect of friction near the coast, mainly due to land mass, is already significant when a quarter of the storm is above the earth. Including these considerations could already improve the aspect of the generated return periods as the intensity at landfall directly depends on the value of κ before landfall. Therefore, inland wind speeds may be overestimated, shifting all the simulated ensemble spreads downwards (reduction of the intensity). However, the magnitude of this shift is not yet evident and could even vary from the different cities as it depends on the averaged storms LMI location. If there is a tendency of high storms' number reaching their LMI far from the coast in a region, then this effect would be all the more noticeable. On the contrary if an area hosts a large number of cyclones that reach their LMI close to land, then the effect of the adjustment would be reduced. The return periods in New Orleans, Miami and Hawaii could be even more underestimated after such an adjustment. On the contrary, it is unlikely that the other return periods would not fall within the range of confidence anymore after the adjustment. Therefore we have to look to the other main factor affecting the shapes of these curves, the $\frac{\text{LMI}}{\text{MPI}}$ ratio, if we want any hope of improving the predictions.

The $\frac{\text{LMI}}{\text{MPI}}$ ratio may even be the most critical factor as it controls the value of the estimated LMI and hence influences the value of the intensity at landfall after decaying. Indeed, the return period curves were seen to be very sensitive to this ratio. In particular, distinguishing the ratio distribution from the storm's LMI distance provided more accurate results. The discrepancy in the goodness of the predictions of the return periods found in the different basins can come from the regional variations of the TCs number. The fact that the West Pacific is the basin with the most significant activity with 22.5 typhoons per year on average [Bloemendaal et al., 2020b] and that a non-negligible amount of $\frac{\text{LMI}}{\text{MPI}}$ data is lacking in the North Atlantic naturally affects the shape of the distribution. Hence, as the distributions' characteristics were seen to vary from the basin (as discussed previously), it could be that the WP distribution is over-represented and the NA and NI ones underrepresented in the overall distribution.

It is hence too early to conclude if a worldwide universal distribution is still appropriate and acceptable for the model's application, namely in making accurate predictions of the cyclones' intensities at landfall. Further work is needed to continue estimating what turns out to be the core of the model's foundation: the $\frac{\text{LMI}}{\text{MPI}}$ distribution being as accurate as possible before including more TC characteristics in the model. Improving the rainfall and wind profile studies should come in a second time, as they depend on the accuracy of the decay parameter and LMI estimations. We hence suggest analysing the ratio distribution again, especially after computing all the lacking MPI values, keeping the distinction of the distribution with the storm's distance from land, as it has shown promising results. Highlighting a distribution weighted continuously by the distance, or

discretising this distribution further, could also provide more accurate return periods. We suggest continuing to search in this way to find an optimal description of the intensification process prior to landfall encoded in the $\frac{LMI}{MPI}$ distribution. After this more profound analysis, if using a universal $\frac{LMI}{MPI}$ distribution does not turn out to describe the return periods curves any better in certain regions, only then making the distinction per basin will be necessary.

It is also relevant to remember that the study of the return periods follows from shifting the trajectory and the initial position of each parent storm, such that the new synthetic storms evolve in different realistic locations. This implementation varies the cyclones' frequency in every studied area. The variation of the TC number encoded in the Poisson distribution also includes an interannual variability in the TC activity. It is hard to quantify the impact of the induced variabilities on the studied locations' return periods that follows from these implementations. However, we are not too worried that if the algorithm deviating the storms shows coherent results on other tests (provided in another research project), the induced TC activity variability should not cause, as a whole, an over or underestimation of the events' frequency. It might be interesting to look at another, more precise, TC interannual variability in a second measure. In particular, on the variability induced by El Niño and La Niña events. These events are part of the El Niño Southern Oscillation (ENSO), which is a coupled ocean-atmosphere phenomenon in the Pacific ocean inducing eastward and westward oscillations in the SST repartition of the ocean. The magnitude of these oscillations were shown to be strongly linked to the worldwide TC activity [Wu et al., 2004, Landsea et al., 2000]. Including the interannual variability in such a manner is also an excellent way to include the variability induced by climate change. In further works, it is thus necessary to explore ENSO indices quantifying the strength of such events and the link with the TC variability.

Finally, The first trend shows that the model well approximates the typhoon's activity in the exposed towns of the West Pacific and North Indian. However, this trend should be confirmed with further works, including more towns and larger scales, to conclude on the goodness of the TCs' wind speeds predictions at landfall in the two basins as a whole. Works remain in studying the effect of the adjusted $\frac{LMI}{MPI}$ and decay parameter distributions on the return periods, especially in the areas not well approximated by the model (NAin particular). If new simulations are planned, especially to extend the return periods on larger timescales, comparing the results with other models generating synthetic cyclones could be interesting. The return period curves derived from the aforementioned STORM model [Bloemendaal et al., 2020a] and the ones computed from [Lee et al., 2018] are examples of results to which our predictions could be compare to. It could also be interesting to change the location scale to compare the return periods obtained per basin in more detail. For example, compare our predictions to the model introduced in [Arthur and Woolf, 2013] modelling cyclone's activity in the Pacific region.

Even though we have predicted 80 additional years based on the past 40 years of historical data so far, and hence the same climate conditions, the relevance of the model appears when this procedure is applied to a complete other set of MPI values, accounting for example for climate change. This study will have to be done hand in hand with the inclusion of a realistic TC interannual variability based on El Niño and la Niña events. The promising results of the return periods at different landfall in the West Pacific and

North Indian oceans to the present climate conditions give confidence in predicting the TC activity change with global warming in these two regions. However, work remains to build a complete model predicting the TC activity in the other basins, such as North Atlantic and East Pacific. It is essential to spend time on obtaining accurate distributions right before landfall as this entirely guides the intensity predictions made by the model. Moreover, the maximum wind speed profile at landfall has not been included in the computations of the return periods. Including the latter will be the next step to provide an even more accurate description of the TC activity at landfall.

5 Conclusion and Outlook

In this report, we presented the foundations of a new numerical model, whose ambition is to become a reference for assessing risks associated with TCs. Unlike the current STORM reference model, it is implemented to be adapted to different climate conditions, in particular in the context of global warming.

To reach our goal, new synthetic storms are generated based on universal cyclone characteristics, highlighted from the past 40 years of historical records (provided by IB-TrACS), and variables which depend on climatic conditions. First, this study focused on extracting the decay parameter κ 's distribution over the ocean as well as over land. This was done by fitting all the storms of the historical dataset to the first order logistical Equation (4) in the two different environments. This distribution provides a worldwide realistic representation of the storms' decay behaviour. Furthermore, to estimate the LMI of each generated storm we extracted the $\frac{\text{LMI}}{\text{MPI}}$ distribution, making, in this first approach, the assumption that this distribution is universal in terms of both season and region. In such a way, as the MPI is sensitive to environmental changes, multiplying the universal ratio with the MPI value provide an accurate estimate of the LMI for a given location and month. Implementing the distribution in such a manner also allows selecting another MPI dataset, accounting for global warming, for example. These considerations, combined with another study taking care of the storm's track deviations, constitute our model's foundation.

Both distributions' dependence on the storm's distance from the coast have been highlighted. This dependence has been implemented for the $\frac{\text{LMI}}{\text{MPI}}$ ratio as three discrete distributions, each accounting for the distribution in three different ranges of the storm's distance from the coast.

Moreover, we tried to include more TCs characteristics in the model, such as its induced precipitations and wind profile, relevant to assess the risk at landfall. In particular, we could not extract any worldwide easily implementable empirical relations between the averaged rain rate induced by a cyclone at landfall and other quantities such as the averaged terrain height, the maximum wind speed or the saturated water vapour pressure. The reasons were discussed, and many suggestions were proposed to continue studying the rain dependencies. This work illustrated the complexity of the TC-induced rain mechanism at landfall. However, we successfully integrated the TC wind profile (averaged azimuthal wind speed as a function of radius from the TC's eye) on land by extracting the power law exponent α and the averaged radius of 34 knots winds R_{34} . Identifying these distributions enabled us to generate a TC's wind footprint at landfall and not only

assess the risk from its maximum wind speed.

Finally, the model's results were compared to the observations by computing the return period curves as a function of landfall in the most exposed cities of different basins. The return periods fell within the ensemble spread of the generated return periods for the studied West Pacific and North Indian cities. However, the model failed to predict the behaviour of TC activity in the cities of the North Atlantic basin. At the same time, the intensity trend in Hawaii was not clear enough to make any conclusion in the goodness of the predictions around it. Many easily implementable recommendations to improve the accuracy of the curves were suggested. I am confident that adjusting the model based on the recommended prescriptions will enhance the prediction's accuracy and could provide a model that lives up to its ambitions.

While not successful in all of its aspects, this report overall demonstrated the relevance of generating new synthetic storms based on a novel procedure to assess the landfall risk. It is hoped that many of the studies in this report will be explored in more detail and further implemented in the model, so it can eventually become useful to the scientific community. More coastal areas will become at risk of TC with climate change [[Emanuel, 2021a](#)] and the past few years hosted some of the more damageable hurricanes with Maria, Irma, and Harvey. Building accurate models assessing risks, like the one suggested here, and further improving it becomes vital and needs to be achieved as fast as possible.

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A Appendix

A.1 Acronyms

- **STORM** - Synthetic Tropical cyclOne geneRation Model
- **IRIS** - Imperial College Storm Model
- **TC** - Tropical Cyclone
- **SST** - Sea Surface Temperature
- **LMI** - Lifetime Maximum Intensity
- **MPI** - Maximum Potential Intensity
- **RMW** - Radius of Maximum Wind
- **MSW** - Maximum Sustained Winds
- **ENSO** - El Niño Southern Oscillation
- **NA** - North Atlantic
- **WP** - West Pacific
- **EP** - East Pacific
- **NI** - North Indian

A.2 Additional informations

The following tables summarise the properties of each fitted distributions. Firstly, Table 2 reviews the properties of the decay parameter κ 's distributions, over land and over the ocean.

Distributions	Log-normal	Beta	Burr
κ_{ocean}	Rejected	Function: $f_{\beta}(\frac{x}{s}; a; b)/s$ Parameters: $a = 1.39, b = 25.10,$ $s = 1.85 \cdot 10^{-6}$	Not studied
κ_{land}	Rejected	Function: $f_{\beta}(\frac{x}{s}; a; b)/s$ Parameters: $a = 2.95, b = 180 \cdot 10^6,$ $s = 33.46$	Not studied

Table 2: Summary of the fitted κ distributions parameters distinguishing between the distribution over land κ_{land} and the distribution over the ocean κ_{ocean} .

Secondly, Table 3 reviews the properties of the $\frac{LMI}{MPI}$ ratio's distributions, distinguishing between every relevant range of distances.

Distributions	Log-normal	Beta	Burr
Worldwide $\frac{LMI}{MPI}$ ratio.	Bimodal function $w_1 f_{\log}(\frac{x-l_1}{s_1}; \mu_1; \sigma_1)/s_1 + w_2 f_{\log}(\frac{x-l_2}{s_2}; \mu_2; \sigma_2)/s_2$ First log-normal parameters f_1 : $\sigma_1 = 0.18$ / $s_1 = \exp \mu_1 = 0.65$ $l_1 = -0.014$ / $w_1 = 0.55$ Second log-normal f_2 : $\sigma_2 = 0.15$ / $s_2 = \exp \mu_2 = 0.96$ $l_2 = -0.02$ / $w_2 = 0.45$	not studied	not studied
Near land $\frac{LMI}{MPI}$ ratio: LMI's location below 300 km	Bimodal function $w_1 f_{\log}(\frac{x-l_1}{s_1}; \mu_1; \sigma_1)/s_1 + w_2 f_{\log}(\frac{x-l_2}{s_2}; \mu_2; \sigma_2)/s_2$ First log-normal f_1 parameters : $\sigma_1 = 0.24$ / $s_1 = \exp \mu_1 = 0.66$ $l_1 = 0.02$ / $w_1 = 0.82$ Second log-normal f_2 : $\sigma_2 = 0.18$ / $s_2 = \exp \mu_2 = 0.99$ $l_2 = 0$ / $w_2 = 0.18$	not studied	not studied
Mid sea $\frac{LMI}{MPI}$ ratio: LMI's location between 300 and 800 km	Bimodal function $w_1 f_{\log}(\frac{x-l_1}{s_1}; \mu_1; \sigma_1)/s_1 + w_2 f_{\log}(\frac{x-l_2}{s_2}; \mu_2; \sigma_2)/s_2$ First log-normal f_1 parameters: $\sigma_1 = 0.16$ / $s_1 = \exp \mu_1 = 0.57$ $l_1 = 0.03$ / $w_1 = 0.53$ Second log-normal f_2 parameters: $\sigma_2 = 0.17$ / $s_2 = \exp \mu_2 = 0.91$ $l_2 = -0.01$ / $w_2 = 0.47$	not studied	not studied
Far sea $\frac{LMI}{MPI}$ ratio: LMI's location over 800 km	Bimodal function $w_1 f_{\log}(\frac{x-l_1}{s_1}; \mu_1; \sigma_1)/s_1 + w_2 f_{\log}(\frac{x-l_2}{s_2}; \mu_2; \sigma_2)/s_2$ First log-normal f_1 parameters: $\sigma_1 = 0.17$ / $s_1 = \exp \mu_1 = 0.64$ $l_1 = 0.01$ / $w_1 = 0.45$ Second log-normal f_2 parameters: $\sigma_2 = 0.12$ / $s_2 = \exp \mu_2 = 0.93$ $l_2 = 0.03$ / $w_2 = 0.55$	not studied	not studied

Table 3: Fitted $\frac{LMI}{MPI}$ distributions' parameters. The characteristics of the worldwide distribution as well as the distribution restricted to different range of the storm's LMI distances are displayed.

Finally, Table 4 reviews the properties of the variables associated to the study of the wind footprint: the decay exponent of the wind profile α and the averaged radius of 34

knots winds R_{34} .

Distributions	Log-normal	Beta	Burr
Exponent α	Rejected	Rejected	Function: $f_{\text{burr}}(\frac{x}{s}; c; d)/s$ Parameters: $c = 4.83$ $d = 0.49$ $s = 0.60$
Averaged R_{34}	Function: $f_{\log}(\frac{x}{s}; \mu; \sigma)/s$ Parameters: $\sigma = 0.49$ $s = \exp \mu = 80.94$	Function: $f_{\beta}(\frac{x-l}{s}; a; b)/s$ Parameters: $a = 4.56$ $b = 5240.63$ $s = 104237.26$ $l = 0.13$	not studied

Table 4: Distributions' parameters associated to the wind footprint study.